The Optimal Strategy in the Sucker Punch Endgame

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Abstract

Many competitive Pokémon situations are essentially small game theory problems. A particularly clean example arises with the priority move *Sucker Punch*, which only succeeds if the opponent attacks. This creates a bluffing interaction between attacking and not attacking.

We model the late–game interaction between *Gholdengo* and *Kingambit* as a stochastic zero–sum game with a finite resource constraint: Sucker Punch has limited PP. Using dynamic programming and mixed strategy equilibrium, we derive the optimal strategies for both players. If Kingambit has k remaining Sucker Punch PP, then the player attacks with probability

$$p_k = \frac{1}{k+1},$$

while Kingambit uses Sucker Punch with probability

$$q_k = \frac{k}{k+1}.$$

Under these strategies, the player's probability of eventual victory is

$$V_k = \frac{1}{k+1}.$$

Monte Carlo simulation closely matches the theoretical prediction and reveals an interesting property: the player is equally likely to win on any possible attack turn.

1 Introduction

Game theory studies strategic decision–making when outcomes depend on the actions of multiple agents. While it is often discussed in economics or politics, the same structures frequently appear in competitive games.

A surprisingly elegant example occurs when facing the priority move *Sucker Punch*. This priority attack only works if the opponent uses an attacking move. Each turn therefore becomes a guessing game.

Consider a late–game interaction between *Gholdengo* and *Kingambit*. Gholdengo can win immediately with *Make It Rain*. However, Kingambit may respond with *Sucker Punch*. Alternatively, Gholdengo might use a non–attacking move such as *Nasty Plot*. If Sucker Punch is used in that case, it fails and consumes one PP.

Because Sucker Punch has only 8 PP, repeated failures gradually weaken Kingambit's threat. The player must therefore balance short–term risk against long–term resource depletion. We analyze this situation using a stochastic game model.

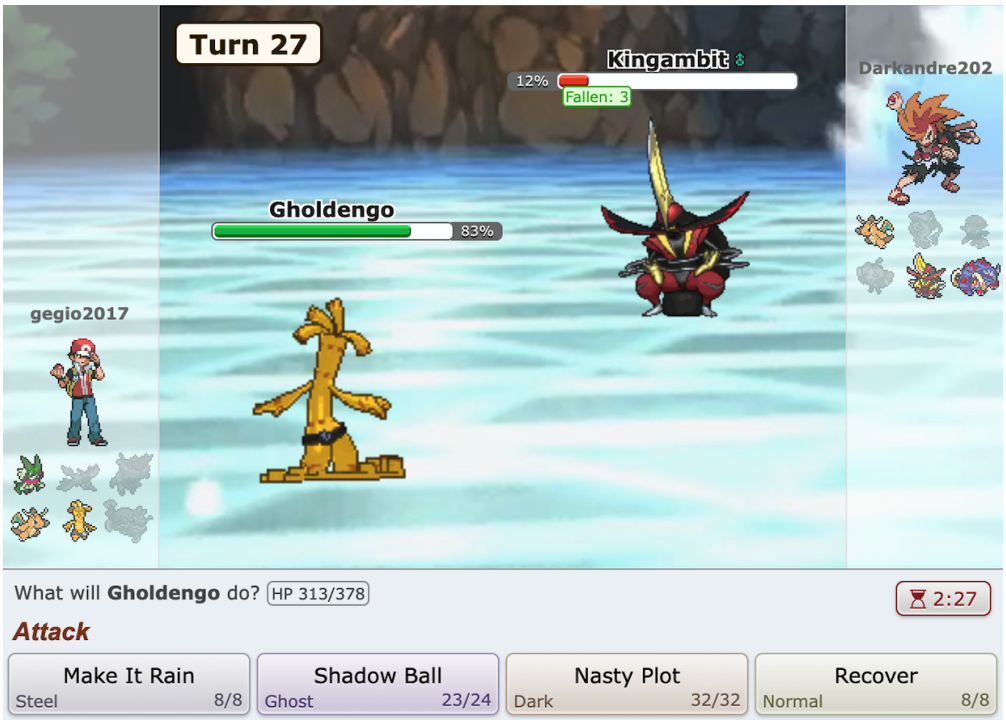

Figure 1: Endgame Scenario with Gholdengo and Kingambit

2 Model

Each turn both players choose an action:

- Player: Attack (A) or Do Not Attack (N)
- Kingambit: Sucker Punch (S) or Regular Attack (K)

Outcomes:

- (A, S): player loses
- (A, K): player wins
- (N, S): Sucker Punch fails and loses one PP
- (N, K): player loses

Let k denote the remaining Sucker Punch PP and define

$$V_k$$

as the probability that the player eventually wins.
The terminal state occurs when Sucker Punch has no PP left:

$$V_0 = 1.$$

Let

$$p = \Pr(\text{player attacks}), \qquad q = \Pr(\text{Kingambit uses Sucker Punch}).$$

The payoff matrix from the player's perspective is

	S	K
A	0	1
N	V_{k-1}	0

so the expected payoff is

$$V_k = p(1-q) + (1-p)qV_{k-1}.$$

The expected payoff V_k can be visualized as a function of the mixed strategies p and q. Figure 2 shows the resulting payoff surface.

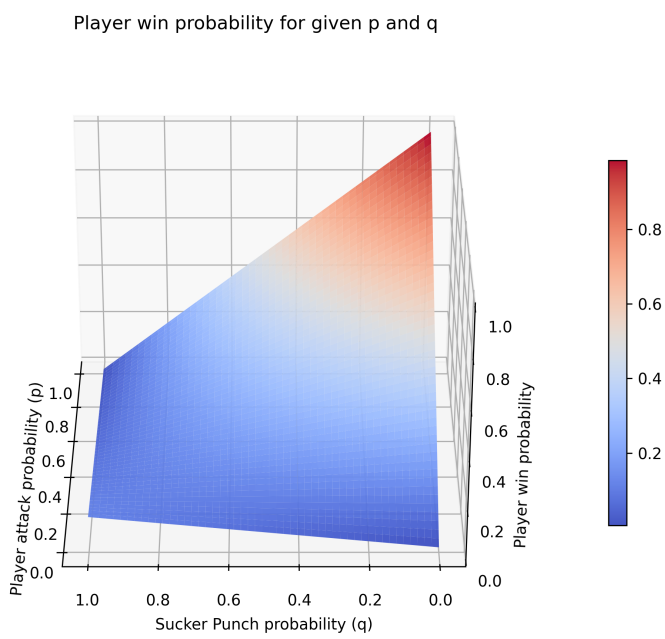

Figure 2: Expected payoff under p and q.

3 Equilibrium Analysis

The game satisfies the Markov property since only the remaining PP matters. Therefore

$$V_k = \max_p \min_q \left[p(1-q) + (1-p)qV_{k-1}\right].$$

No pure strategy equilibrium exists, so both players must mix.

3.1 Player Indifference

If the player attacks, the expected payoff is

$$1 - q.$$

If the player does not attack, the payoff is

$$qV_{k-1}.$$

Equating them,

$$1 - q = qV_{k-1}$$

which gives

$$q = \frac{1}{1 + V_{k-1}}.$$

3.2 Kingambit Indifference

If Kingambit uses Sucker Punch, the player's payoff is

$$(1 - p)V_{k-1}.$$

If Kingambit attacks normally, the payoff is

$$p.$$

Equating them,

$$p = (1 - p)V_{k-1}$$

which yields

$$p = \frac{V_{k-1}}{1 + V_{k-1}}.$$

3.3 Value Recursion

Substituting these equilibrium probabilities into the value expression,

$$V_k = p(1 - q) + (1 - p)qV_{k-1},$$

we obtain

$$V_k = \frac{V_{k-1}^2}{(1 + V_{k-1})^2} + \frac{V_{k-1}}{(1 + V_{k-1})^2} = \frac{V_{k-1}}{1 + V_{k-1}}.$$

4 Closed Form Solution

Theorem 1. *If Kingambit has k remaining Sucker Punch PP, the player's probability of eventual victory is*

$$V_k = \frac{1}{k + 1}.$$

The equilibrium strategies are

$$p_k = \frac{1}{k + 1}, \qquad q_k = \frac{k}{k + 1}.$$

Proof. The base case is $k = 0$:

$$V_0 = 1 = \frac{1}{1}.$$

Assume

$$V_{k-1} = \frac{1}{k}.$$

Using the recursion,

$$V_k = \frac{V_{k-1}}{1 + V_{k-1}} = \frac{\frac{1}{k}}{1 + \frac{1}{k}} = \frac{1}{k+1}.$$

Thus the formula holds by induction.

Substituting $V_{k-1} = 1/k$ into the indifference conditions gives

$$p_k = \frac{1}{k+1}, \qquad q_k = \frac{k}{k+1}.$$

□

5 Dynamic Programming Results

The recursion

$$V_k = \frac{V_{k-1}}{1 + V_{k-1}}$$

together with the indifference conditions

$$p_k = \frac{V_{k-1}}{1 + V_{k-1}}, \qquad q_k = \frac{1}{1 + V_{k-1}}$$

allows the equilibrium values to be computed iteratively from the base case $V_0 = 1$.

Table 1 reports the win probability, attack probability, and Sucker Punch probability for each remaining PP state. These values match the closed-form solution derived above:

$$V_k = p_k = \frac{1}{k+1}, \qquad q_k = \frac{k}{k+1}.$$

PP	Win Probability V_k	Attack Probability p_k	Sucker Punch Probability q_k
0	1.000	1.000	0.000
1	0.500	0.500	0.500
2	0.333	0.333	0.667
3	0.250	0.250	0.750
4	0.200	0.200	0.800
5	0.167	0.167	0.833
6	0.143	0.143	0.857
7	0.125	0.125	0.875
8	0.111	0.111	0.889

Table 1: Dynamic programming results for each remaining Sucker Punch PP state.

6 Simulation

To verify the model, we simulated 100,000 games starting from $k = 8$. Both players followed the equilibrium strategies.

The empirical win rate was

$$0.11107,$$

which closely matches the theoretical value

$$V_8 = \frac{1}{9}.$$

The turn on which the winning attack occurred was also recorded.

Winning Turn	Number of Wins
1	1220
2	1269
3	1244
4	1191
5	1214
6	1253
7	1254
8	1241
9	1221

Table 2: Winning attack turns from 100,000 simulations.

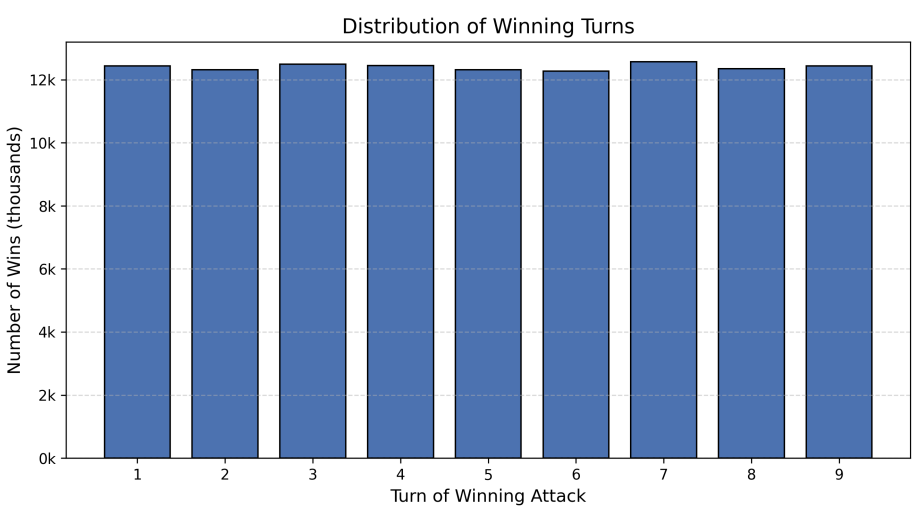

Figure 3: Distribution of winning attack turns.

7 Discussion

A striking feature of the simulation is that wins are almost perfectly uniform across turns. This follows directly from the equilibrium structure.

At state k the probability of an immediate attack–win is

$$p_k(1 - q_k) = \frac{1}{(k+1)^2}.$$

The probability of surviving a failed Sucker Punch is

$$(1 - p_k)q_k = \frac{k^2}{(k+1)^2}.$$

Suppose the game begins with K remaining PP. Winning on turn t requires surviving $t - 1$ failures and then attacking successfully:

$$\Pr(\text{win on turn } t) = \left(\prod_{j=0}^{t-2} \frac{(K-j)^2}{(K-j+1)^2} \right) \frac{1}{(K-t+2)^2}.$$

Expanding the product,

$$\frac{K^2}{(K+1)^2} \cdot \frac{(K-1)^2}{K^2} \cdots \frac{(K-t+2)^2}{(K-t+3)^2} \cdot \frac{1}{(K-t+2)^2}.$$

Everything cancels telescopically, leaving

$$\Pr(\text{win on turn } t) = \frac{1}{(K+1)^2}.$$

Thus every possible winning turn has exactly the same probability.

For our simulation with $K = 8$,

$$\Pr(\text{win on turn } t) = \frac{1}{(8+1)^2} = \frac{1}{81}.$$

Since there are 9 possible winning turns, the total probability of eventual victory is

$$9 \cdot \frac{1}{81} = \frac{1}{9},$$

which matches the theoretical value $V_8 = 1/9$.

8 Limitations

This analysis assumes both players follow the optimal mixed strategies. In practice, real players rarely play perfectly optimally and may overuse Sucker Punch, hesitate to attack, or develop predictable patterns. The equilibrium derived here should therefore be interpreted as a benchmark for optimal play rather than a literal description of typical gameplay.

9 Conclusion

We modeled the Sucker Punch endgame as a stochastic zero–sum game with a finite resource constraint. Using equilibrium analysis we showed that If Kingambit has k remaining Sucker Punch PP, the player's win probability is

$$V_k = \frac{1}{k+1},$$

and the optimal attack probability is

$$p_k = \frac{1}{k+1}.$$

Simulation confirms the theoretical prediction and reveals an elegant property: under optimal play, every possible winning attack turn occurs with equal probability.

This example shows how classical game theory can illuminate strategy in competitive games, especially when bluffing and limited resources are both involved.